

Multiplication

$$X = 5281$$

$$Y = 23492$$

$\text{length}(X) = m$
 $\text{length}(Y) = n$

		2							
		5	2	8	1				
X	2	3	4	9	2				
		1	0	5	6	2			
	4	7	5	2	9				

$2 \cdot 5281$
 $30 \cdot 5281$
 $400 \cdot 5281$
 $\vdots$

$+$

$\uparrow \uparrow \uparrow$

gradeschool multiplication runtime

$$\Theta(m \cdot n)$$

for all m, n

best / worst case?

same!

but answer is

$$5281 \cdot 10000$$

$$= 52810000$$

what are m, n ? as function of x, y

better algs?

better alg?

not better alg?

wasn't been proven?

Recursive Multiplication

$x = \begin{array}{|c|c|c|c|c|} \hline 2 & 8 & 0 & 8 & 8 & 9 & 9 & 6 \\ \hline \end{array}$ $n = 8$
 $y = \begin{array}{|c|c|c|c|c|} \hline 2 & 6 & 7 & 1 & 5 & 6 & 3 & 5 \\ \hline \end{array}$

$\text{multiply}(x, y, n) :$

if $n = 1 :$

return $x \cdot y$

if $n > 1 :$

call $\text{multiply}(_, _, m)$

use result to get $x \cdot y$

$m < n$

$\downarrow$